



UNIVERSITÀ DI PISA

# **Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing**

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**Candidato:**

Alessandro Gori

**Relatore:**

Prof. Giacomo Mazza

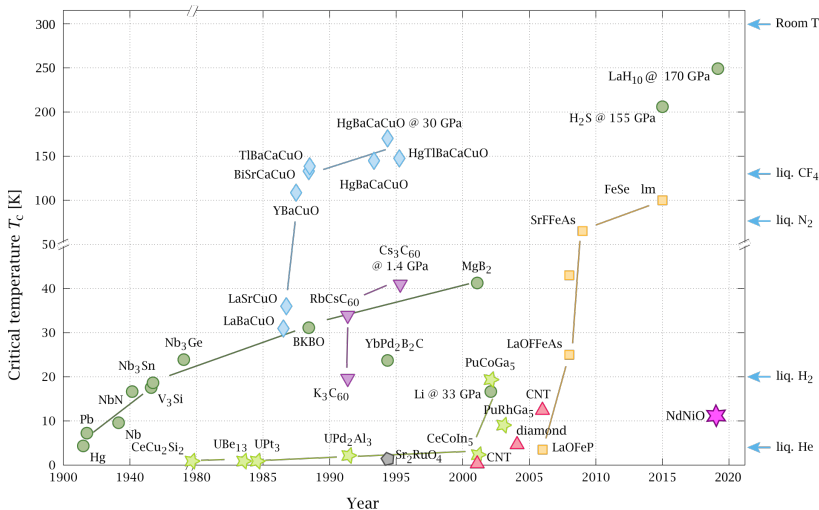
# Outline

1. Cuprates and Hubbard physics
2. The Hartree-Fock method
3. The normal phase
4. The antiferromagnetic phase
5. The superconducting phase
6. Conclusions and outlook

## Cuprates and Hubbard physics

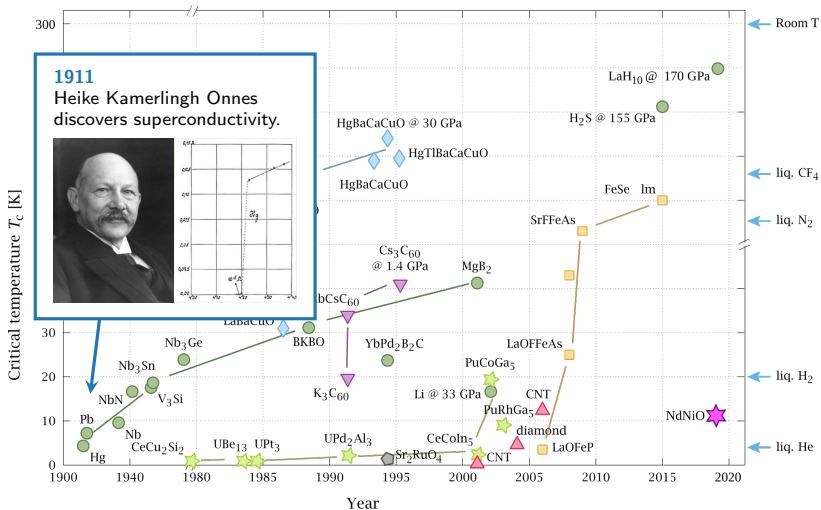
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# The timeline of superconductivity



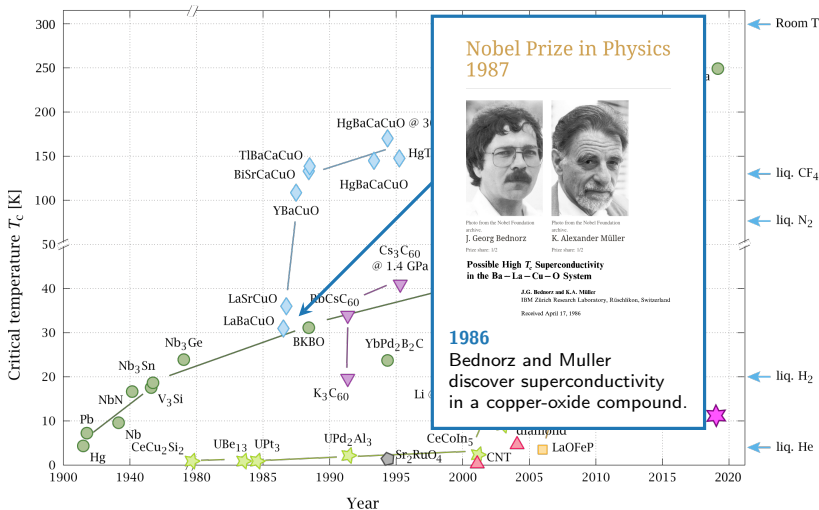
**Figure 1:** Timeline of the discovery of superconductivity in various compounds. The cuprates branch is represented as cyan diamonds. Image source: P. Ray and O. Gingras, distributed according to [the license CC BY-SA 4.0](https://creativecommons.org/licenses/by-sa/4.0/).

# The timeline of superconductivity



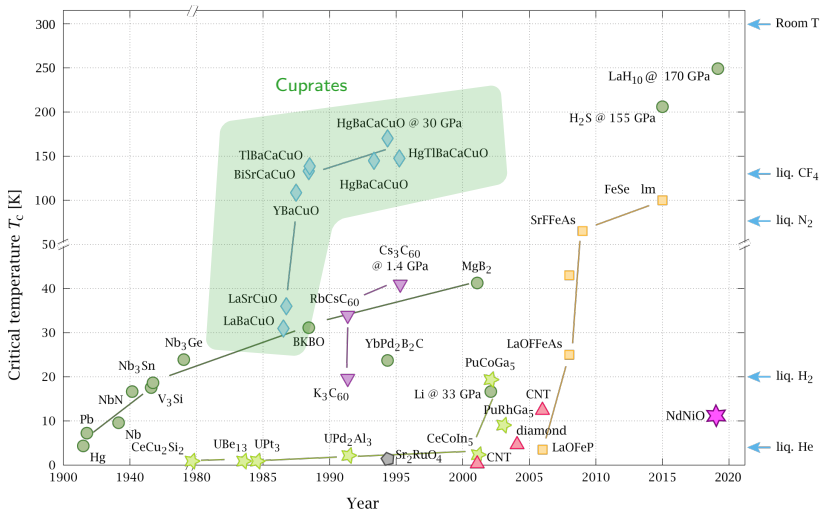
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**Figure 1:** Timeline of the discovery of superconductivity in various compounds. The cuprates branch is represented as cyan diamonds. Image source: P. Ray and O. Gingras, distributed according to the license CC BY-SA 4.0.

# The timeline of superconductivity



**Figure 1:** Timeline of the discovery of superconductivity in various compounds. The cuprates branch is represented as cyan diamonds. Image source: P. Ray and O. Gingras, distributed according to [the license CC BY-SA 4.0](#).

# The phase diagram of cuprates

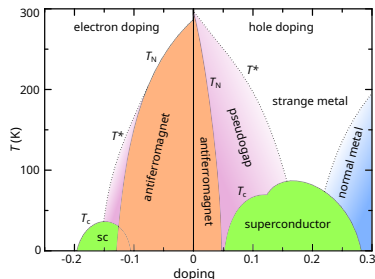
Cuprates are the prototypical example of **strongly correlated materials**:

- ◇ **Antiferromagnetism** at null and low doping;
- ◇ ***d*-wave superconducting** domes.

Both arise from **strong electronic correlations**.



Failure of band theory in microscopic description.

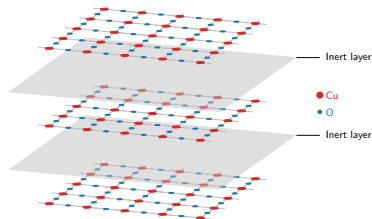


Schematic phase diagram of cuprates. Author: Holger Motzkau, [Wikimedia commons](#), distributed according to [the license CC BY-SA 4.0](#)



These materials are stackings of:

- ◇ 2D copper oxide active layers;
- ◇ intermediate layers of inert material.



## The single-band Hubbard model on the square lattice

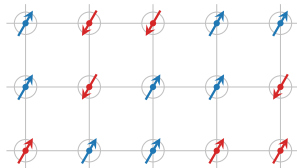
$$\hat{H}_{\text{HM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{On-site repulsion}} \quad \text{with } t, U > 0.$$

**What is the origin of the insulation mechanism?**

The strongly coupled Hubbard model,

$$t \ll U,$$

can describe the **Mott insulator** state.

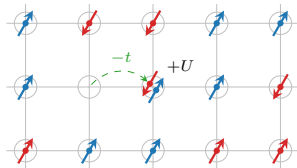


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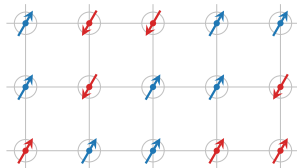
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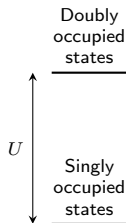


## Mott insulation mechanism

Electrons are **localised** as in a sort of “traffic jam”.

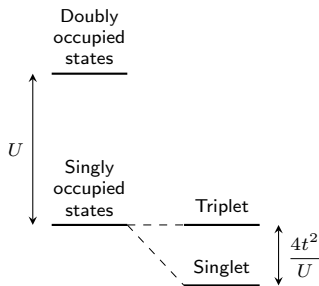


On the 2-sites Hubbard model with 2 electrons, at  $U \gg t$ , singlet pairing is energetically favoured:



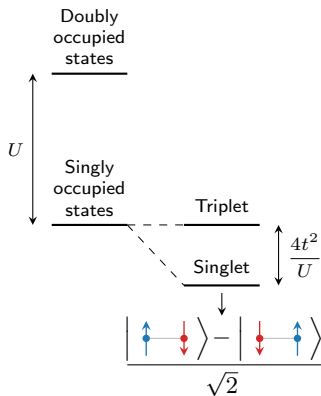
## Antiferromagnetism in the undoped Hubbard model

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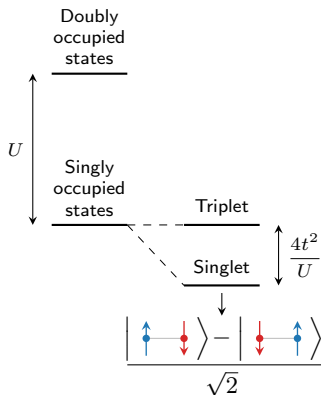
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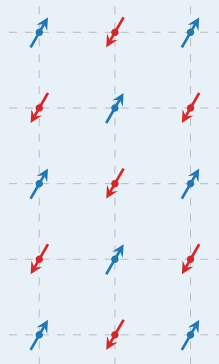
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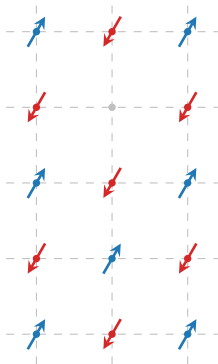
## Extension to $N$ sites

On the square lattice, AF ordering is favourite:



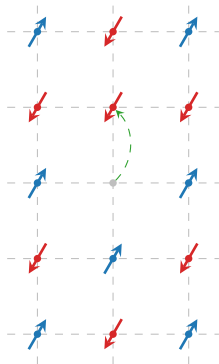


How does superconductivity emerge from doping the AF state in the Hubbard model?



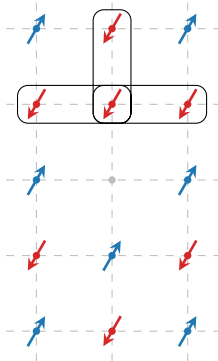
Superconductivity emerges from attractive interaction between electrons.

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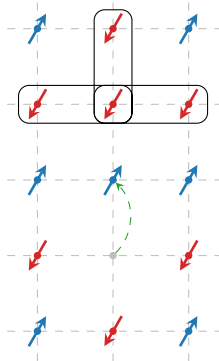
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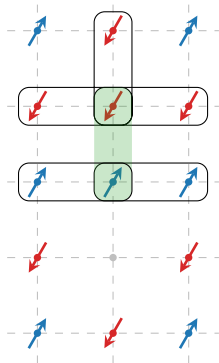
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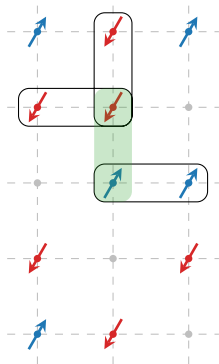
Superconductivity emerges from attractive interaction between electrons.

Neighbouring electrons with opposite spin  
“move together”:



**Effective** attractive interaction.

How does superconductivity emerge from doping the AF state in the Hubbard model?



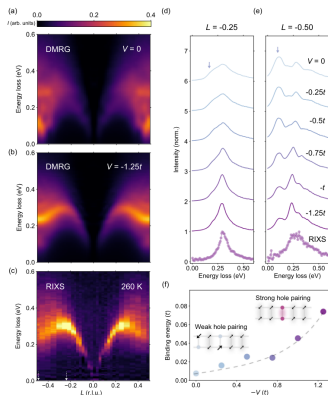
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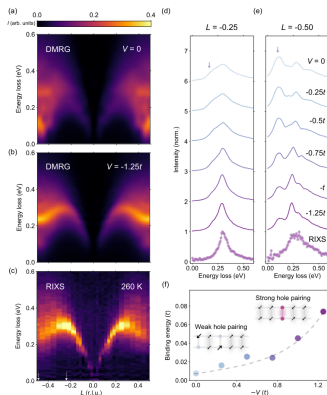
# Recent developments: the extended Hubbard model



**Observed:** incompatibilities of the Hubbard model with the spectrum of magnetic excitation in certain cuprates.

Hari Padma et al., 2025. *Physical Review X* 15 (2): 021049.

# Recent developments: the extended Hubbard model



Hari Padma et al., 2025. *Physical Review X* 15 (2): 021049.

**Observed:** incompatibilities of the Hubbard model with the spectrum of magnetic excitation in certain cuprates.



Proposal: extension of the model with a **non-local attraction**,

$$\hat{H}_{\text{EHM}} = \hat{H}_{\text{HM}} - V \underbrace{\sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\hat{H}_V}.$$

$\hat{H}_V$  can act as an independent source of  $d$ -wave superconductivity.



**The model (EHM):**

$$\hat{H}_{\text{EHM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{Local repulsion}} - \underbrace{V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\text{Non-local attraction}} \quad \text{with } t, U, V > 0.$$

**The motivation:** compatibility with some aspects of cuprates physics, including  $d$ -wave superconductivity.

**The target:** understand how the new interaction relates to Mott physics, antiferromagnetism and superconductivity.

**The method:** Hartree-Fock variational method, at low temperature.

## The Hartree-Fock method

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# The constrained minimisation framework

We want to approximate the true ground state  $|\Psi_0\rangle$  of the EHM, with energy  $E_0$ .

**Variational principle:**

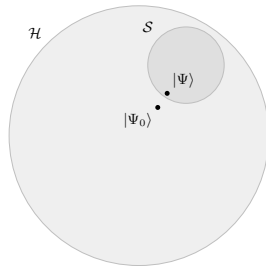
$$\forall |\Phi\rangle \in \mathcal{S} \subset \mathcal{H} : E[\Phi] \geq E_0.$$

$\Downarrow$

## Constrained minimisation procedure

$$|\Psi\rangle : \left. \frac{\delta E[\Phi]}{\delta |\Phi\rangle} \right|_{|\Phi\rangle=|\Psi\rangle} = 0.$$

The state  $|\Psi\rangle$  is the best approximation to  $|\Psi_0\rangle$  inside of  $\mathcal{S}$ .



Sketch of the Hilbert space  $\mathcal{H}$  and a subset  $\mathcal{S}$ .

## The Hartree-Fock subspace of states

$\mathcal{S}$  is the set of all **gaussian states**,

$$|\Psi\rangle = \bigotimes_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} |\Omega_{\gamma}\rangle, \quad \left\{ \hat{\gamma}_{\alpha}, \hat{\gamma}_{\beta}^{\dagger} \right\} = \delta_{\alpha\beta},$$

with  $\hat{\gamma} \leftrightarrow \hat{c}$  through a unitary map.

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with  $\hat{\gamma} \leftrightarrow \hat{c}$  through a unitary map.



Each gaussian state can be identified **uniquely** by fixing

$$\Delta_{\alpha\beta}^{+-} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta} \rangle, \quad \Delta_{\alpha\beta}^{++} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle.$$

These are the **variational parameters** collected in the vector  $\mathbf{v}$ . We aim to find the optimal  $\mathbf{v}$ , such that

$$|\Psi[\mathbf{v}]\rangle \simeq |\Psi_0\rangle.$$

**Phase transition:** spontaneous breaking of some symmetries of the model.

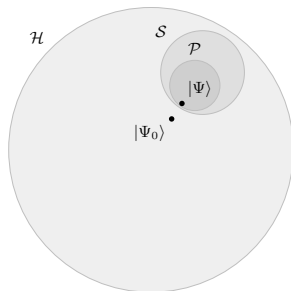
Phase  $X$ : shows symmetries  $Y, Z \dots$



Restrict to  $\mathcal{P} \subset \mathcal{S}$ , gaussian states with symmetries  $Y, Z \dots$



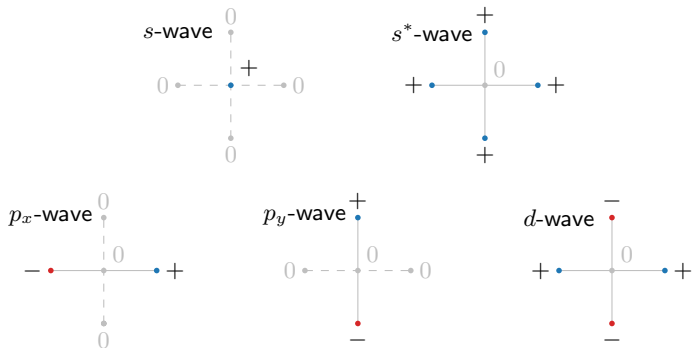
Symmetry selection rules on  $\mathbf{v}$  entries.



Identify symmetries  $\rightarrow$  Derive  $\mathbf{v}$   $\rightarrow$  Solve the variational problem.

## The spatial symmetry channels

It is useful to introduce five spatial symmetry channels:



Function  $f$  on NNs: decomposition in harmonics  $f^{(\gamma)}$ .

## The normal phase

---



## The variational problem for the normal phase

**Normal phase:** no broken symmetries. Just one variational parameter,

$$\mathbf{v} = \left[ u^{(s^*)} \right],$$

where  $u^{(s^*)}$  is the  $s^*$ -wave component of  $u_{\mathbf{k}\sigma} \equiv \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle$ .

$\Downarrow$

### Renormalisation of kinetics

Hopping amplitude is **isotropically renormalised**,

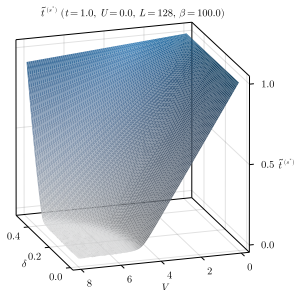
$$t \rightarrow \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2},$$

with the **theoretical constraint**:

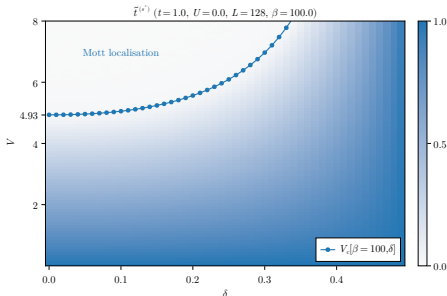
$$0 < \tilde{t}^{(s^*)} < t \quad \implies \quad \text{Possible Mott localisation.}$$

## Complete flattening of bare bands

In  $(\delta, V)$  plane, we have determined theoretically the presence of a Mott critical boundary  $V_c[\beta, \delta]$ ,



(a) Numerical results for  $\tilde{t}^{(*)}$ .



(b) Critical Mott boundary.

Within the Mott plateau, electrons are completely localised and **bands are flattened**.

The role of  $V/t$ :

- ◇ **Renormalise hopping;**
- ◇ **Induce complete Mott localisation:**

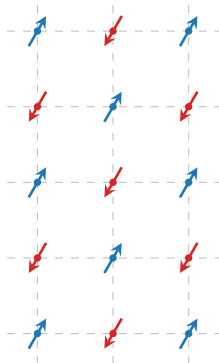
Effective kinetics energy scale  $\ll$  Interaction energy scale .

The role of  $U/t$ :

- ◇ At HF level,  $U$  does not produce Mott localisation.

## The antiferromagnetic phase

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In cuprates, the AF phase is **commensurate**:

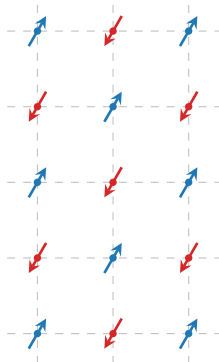
- ◇ 2-sites periodic in each direction;
- ◇ Sublattices oppositely magnetised;
- ◇ Zero net magnetisation.



**Staggered magnetisation  $m$ :**

$$2m = |\langle \hat{n}_{i\uparrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle|,$$

with  $i$  a site.



In cuprates, the AF phase is **commensurate**:

- ◇ 2-sites periodic in each direction;
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**Staggered magnetisation**  $m$ :

$$2m = |\langle \hat{n}_{i\uparrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle|,$$

with  $i$  a site.

### Half-filling restriction

We limit to  $\delta = 0$  (half-filling) due to **stability considerations**.

**Variational parameters** for the AF phase:

$$\mathbf{v} = \begin{bmatrix} m \\ u^{(s^*)} \end{bmatrix},$$

with  $u^{(s^*)}$  defined as for the normal phase.

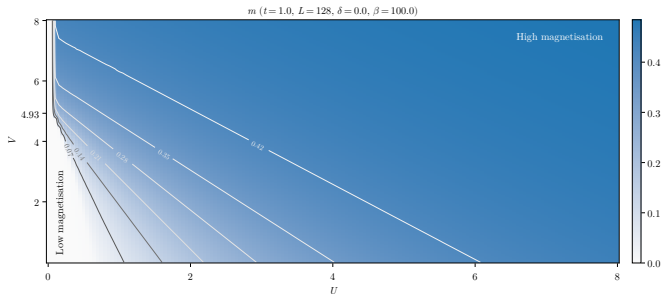
### Renormalisation of kinetics

Mott localisation effect **formally** identical to the normal phase:

$$t \rightarrow \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2},$$

with the **theoretical constraint**:  $0 < \tilde{t}^{(s^*)} < t$ .

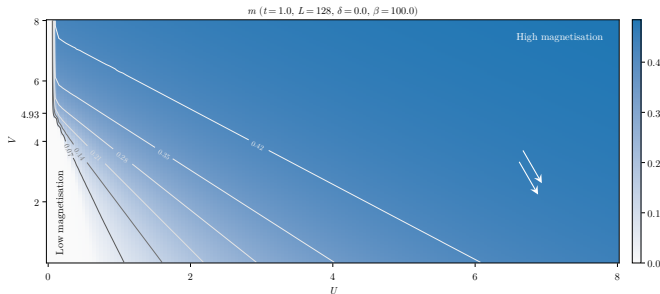
# The cooperation on magnetisation



**Figure 4:** Numerical results for  $m$ . The gray curves indicate isovalued lines.



# The cooperation on magnetisation

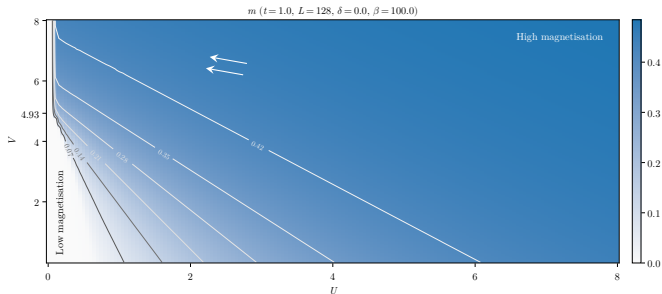


**Figure 4:** Numerical results for  $m$ . The gray curves indicate isovalued lines.

$$U \gg V, t$$

“Conventional” Hubbard magnetisation, induced by the ratio  $U/t$ .

# The cooperation on magnetisation

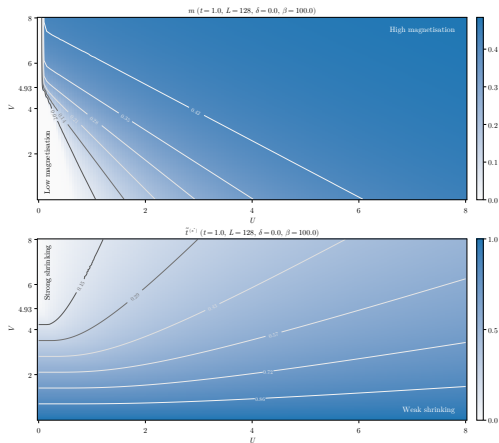


**Figure 4:** Numerical results for  $m$ . The gray curves indicate isovalued lines.

$$V \gg U, t$$

New “unconventional” source of magnetisation, induced by the ratio  $V/t$ .

# The pairing enhancement of magnetisation



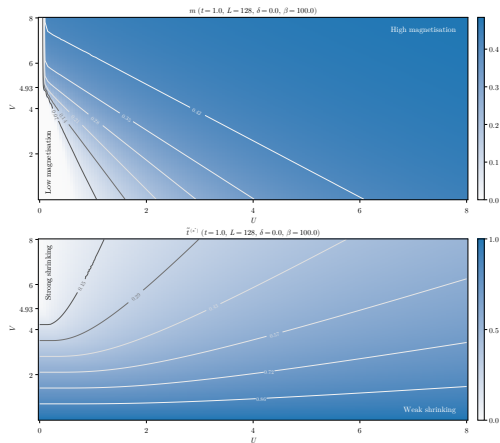
At low temperature:

$$m \simeq \frac{1}{2} + \mathcal{O}(r^2),$$

where  $r = \tilde{t}^{(s^*)}/U$ .

**Figure 5:** Numerical results for  $m$  (top) and  $\tilde{t}^{(s^*)}$  (bottom). The gray curves indicate isovalued lines.

# The pairing enhancement of magnetisation



At low temperature:

$$m \simeq \frac{1}{2} + \mathcal{O}(r^2),$$

where  $r = \tilde{t}^{(s^*)}/U$ .



To enhance  $m$ , either:

- ◇ Increase  $U$ ;
- ◇ **Decrease  $\tilde{t}^{(s^*)}$** .

**Figure 5:** Numerical results for  $m$  (top) and  $\tilde{t}^{(s^*)}$  (bottom). The gray curves indicate isovalued lines.

The role of  $V/t$ :

- ◇ **Renormalise hopping;**
- ◇ **Magnetise the lattice using Mott physics;**

The role of  $U/t$ :

- ◇ Magnetise the lattice, conventional Hubbard physics.



**Cooperation** among interactions to magnetise the lattice.

## The superconducting phase

---

**SC phase:** breaking of charge conservation.

Cooper pairing  $\implies$  SC gap, roughly the “binding energy” of the pairs.

### Gap symmetries and Cooper pairing channels

The gap is a function of momentum with harmonics  $\tilde{\Delta}^{(\gamma)}$ .

- ◇ **Singlet channel:** symmetric components,

$$\tilde{\Delta}^{(s)}, \quad \tilde{\Delta}^{(s^*)} \quad \text{and} \quad \tilde{\Delta}^{(d)}.$$

- ◇ **Triplet channel:** antisymmetric components,

$$\tilde{\Delta}^{(p_x)} \quad \text{and} \quad \tilde{\Delta}^{(p_y)}.$$

We limit to the **singlet channel**. Let the quantities:

$$u_{\mathbf{k}} \equiv \left\langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \right\rangle, \quad w_{\mathbf{k}} \equiv \left\langle \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{-\mathbf{k}\downarrow}^\dagger \right\rangle.$$

**Variational parameters:** symmetric harmonics,

$$\mathbf{v} = \left( u^{(s^*)}, u^{(d)}, w^{(s)}, w^{(s^*)}, w^{(d)} \right)$$

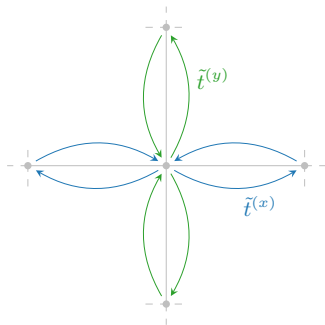
### The role of each parameter

- ◇  $u^{(s^*)}, u^{(d)}$ : renormalisation of kinetics;
- ◇  $w^{(s)}, w^{(s^*)}$  and  $w^{(d)}$ : components of the SC gap.



In the SC phase, the renormalisation of  $t$  gives two components,

$$t \rightarrow \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2} \quad \text{and} \quad \tilde{t}^{(d)} \equiv -\frac{u^{(d)}V}{2}.$$



### Anisotropic hopping

Direction-resolved effective hopping:

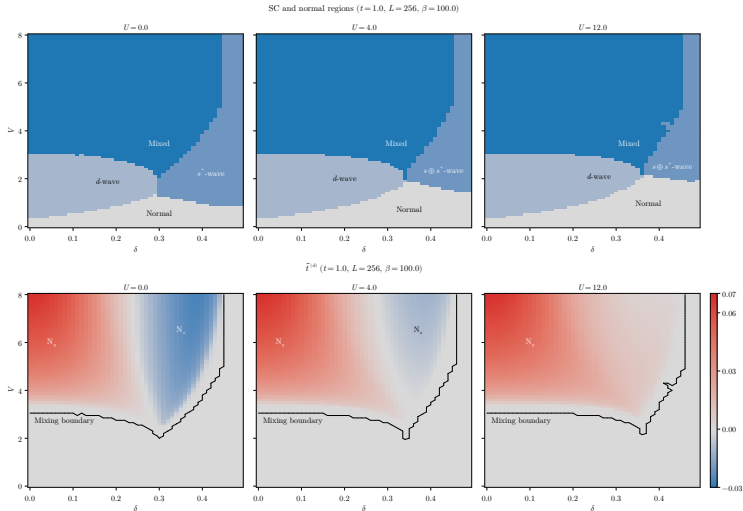
$$\tilde{t}^{(x)} \equiv \frac{\tilde{t}^{(s^*)} + \tilde{t}^{(d)}}{2},$$

$$\tilde{t}^{(y)} \equiv \frac{\tilde{t}^{(s^*)} - \tilde{t}^{(d)}}{2}.$$

$\Downarrow$

$\tilde{t}^{(d)} \neq 0 \implies$  **nematic SSB.**

# Symmetry mixing and insurgence of nematicity



**Figure 6:** Comparison of phase diagrams for the singlet SC phase and convergence values for the nematic order parameter.

The role of  $V/t$ :

- ◇ **Renormalise hopping anisotropically;**
- ◇ **Induce mixed-symmetry superconductivity;**
- ◇ **Induce nematic symmetry breaking.**

The role of  $U/t$ :

- ◇ **Participate in superconductivity even if locally repulsive.**

## Conclusions and outlook

---

**The target:** understand how the new interaction relates to Mott physics, antiferromagnetism and superconductivity.

At HF level, the non-local attraction:

- ◇ Acts as a source of Mott localisation;
- ◇ Enhances magnetisation at half filling;
- ◇ Can produce mixed-symmetry superconductivity;
- ◇ Can produce nematic superconductivity (unexpected, but observed!).

From here, we can follow various paths:

1. **Same model, other phases.**

Triplet superconductivity, breaking of translational invariance, nematic antiferromagnetism. . .

2. **Expand the model.**

Second nearest neighbours hopping, more refined potentials. . .

3. **More advanced numerical techniques.**

DMRG for chains and ladders; hybrid DMFT (local sector) and HF (non-local sector) to retrieve Mott physics induced by  $U$ . . .

**That's all, thank you!**

## Backup 1: details on Mott localisation

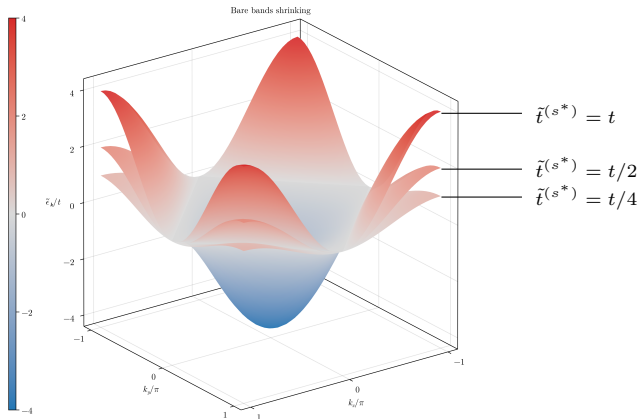
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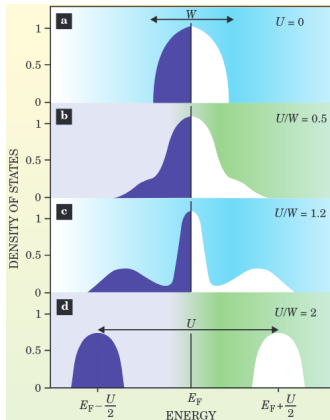
## Bands flattening: a signature of Mott localisation (1/2)

Renormalisation of electronic bands,  $\epsilon_{\mathbf{k}\sigma} \rightarrow \tilde{\epsilon}_{\mathbf{k}\sigma}$ :

$0 < \tilde{t}^{(s^*)} < t \implies$  **Flattening** of bands = Mott localisation.



## Bands flattening: a signature of Mott localisation (2/2)



Gabriel Kotliar and Dieter Vollhardt, 2004.  
*Physics Today* 57 (March): 53–59.

As we increase the ratio  $U/t$ , the many-body DoS undergoes a transition:

- (a) Free model, conventional DoS;
- (b) Weak coupling, large coherent peak;
- (c) Intermediate coupling, **narrow coherent peak**;
- (d) Strong coupling, lower and upper Hubbard bands.



Localisation = flattening of bare bands.

## Backup 2: details on the HF method

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**Hartree-Fock method:**  $\mathcal{S}$  is the set of all **gaussian states**, i.e. states that are non-interacting for some fermionic operators

$$|\Psi\rangle = \bigotimes_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} |\Omega_{\gamma}\rangle, \quad \{\hat{\gamma}_{\alpha}, \hat{\gamma}_{\beta}^{\dagger}\} = \delta_{\alpha\beta},$$

with  $\hat{\gamma} \leftrightarrow \hat{c}$  through a unitary map.



**Wick's theorem:** averages of quartic operators decompose nicely,

$$\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \hat{c}_{\delta} \rangle = \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\delta} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\gamma} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\delta} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle \langle \hat{c}_{\gamma} \hat{c}_{\delta} \rangle}_{\text{Cooper}}.$$

The EHM hamiltonian is a quartic operator!

Each  $|\Psi\rangle \in \mathcal{S}$  is uniquely defined by a particular choice of the **variational parameters** (HFPs)

$$\Delta_{\alpha\beta}^{+-} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta} \rangle, \quad \Delta_{\alpha\beta}^{++} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle.$$

Let  $\mathbf{v}$  be the vector that collects all HFPs. We aim to find

$$|\Psi[\mathbf{v}]\rangle \simeq |\Psi_0\rangle,$$

with  $\mathbf{v}$  such that:

### Variational picture

Energy is minimised,

$$\left. \frac{\delta E[\Phi]}{\delta |\Phi\rangle} \right|_{|\Phi\rangle=|\Psi[\mathbf{v}]\rangle} = 0.$$

### Self-consistency picture

A self-consistency problem is satisfied,

$$A(\mathbf{v}) = \mathbf{v},$$

being  $A$  a non-linear map **uniquely determined**.

# The iterative algorithm (iHF)

The HF solution is the **fixed point** of a non-linear map  $A$ ,

$$A(\mathbf{v}) = \mathbf{v}.$$



**Strategy:** “follow” the map up to the fixed point,

$$\mathbf{v}_{i+1} \equiv gA(\mathbf{v}_i) + (1 - g)\mathbf{v}_i.$$

with  $i > 0$ ,  $g \in [0, 1]$ .

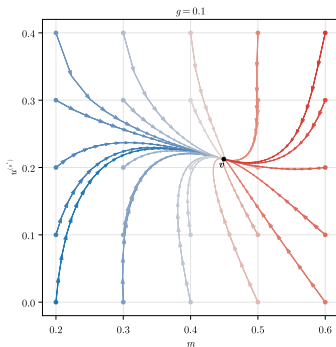


**Convergence criterion:**

$$\forall j \quad : \quad [\mathbf{v}_{i+1} - \mathbf{v}_i]_j < \Delta v_j,$$

with  $\Delta \mathbf{v}$  the vector of tolerances.

Example path (AF) ( $t = 1.0$ ,  $U = 3.0$ ,  $V = 6.0$ ,  $L = 128$ ,  $\beta = 100.0$ ,  $\delta = 0.0$ )



Example path for the AF phase.